

Question 1

You decide to create a Naïve Bayes classifier to distinguish Zombies from AUS students.

You survey Zombies and Healthy AUS students based on the following true and false questions:

W – Do you wear tattered clothes?

S – Do you get very little sleep?

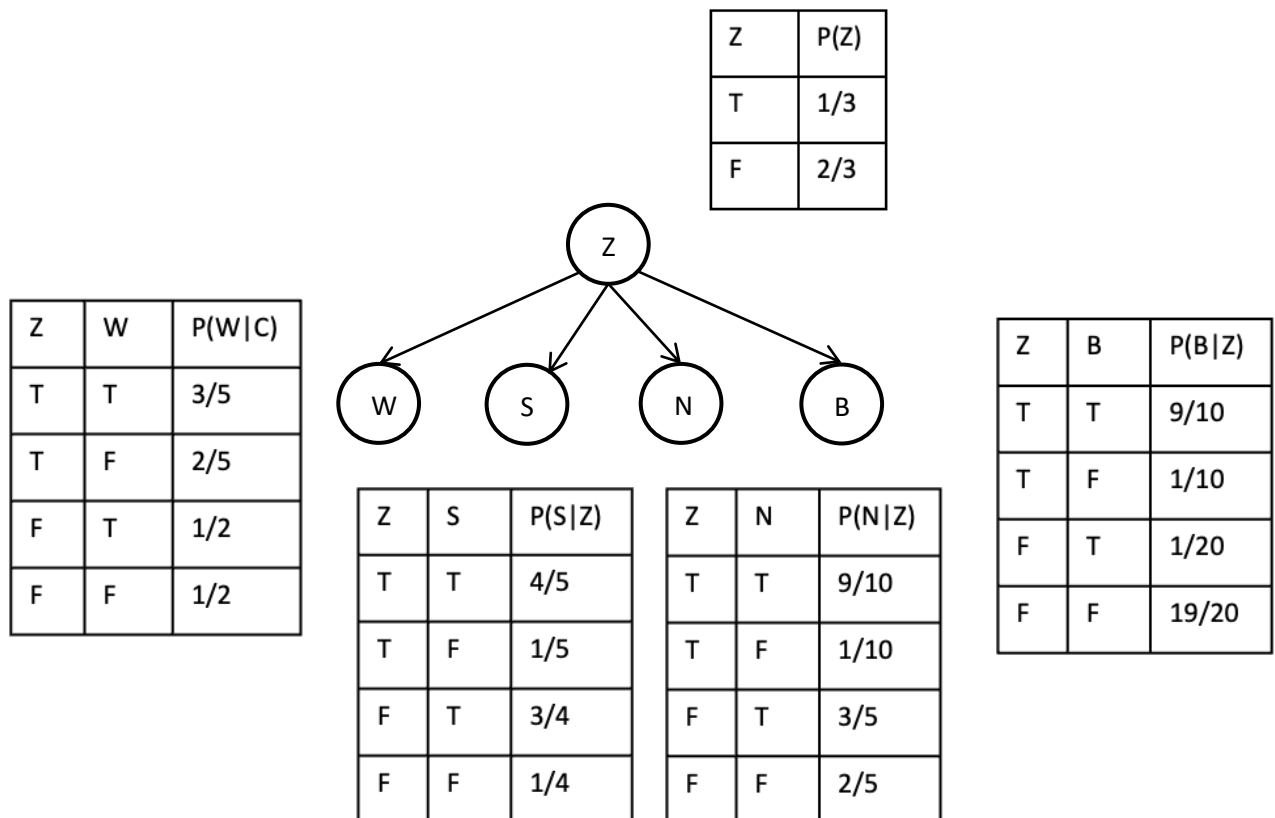
N – Do you often venture out at night?

B – Do you enjoy a healthy diet of human brains?

Your survey returned the following data set:

	No of students with W=True	No of students with S=True	No of students with N=True	No of students with B=True	No of students Surveyed
Zombie	6	8	9	9	10
Non-zombie	10	15	12	1	20

Fill the Conditional Probability Tables (CPT) for the Naïve Bayes Classifier



How do you write the full join distribution $P(Z,W,S,N,B)$?

$$P(Z)P(W|Z)P(S|Z)P(N|Z)P(B|Z)$$

You decide to test your classifier on your friend. He has the following characteristic:

He is a night owl (likes to go out late into the night). He wears old and tattered clothes, and because of his classes he gets very little sleep. He would never ever eat human brains.

To get full credit, you must show your calculations:

$$P(Z|W=T,S=T,N=T,B=F) = \alpha P(Z)P(W=T|Z) P(S=T|Z) P(N=T|Z) P(B=F|Z)$$

$$P(Z=T|W=T,S=T,N=T,B=F) = \alpha \frac{1}{3} * \frac{3}{5} * \frac{4}{5} * \frac{9}{10} * \frac{1}{10} = \alpha * 0.0144$$

$$P(Z=F|W=T,S=T,N=T,B=F) = \alpha \frac{2}{3} * \frac{1}{2} * \frac{3}{4} * \frac{3}{5} * \frac{19}{20} = \alpha * 0.1425$$

$$\alpha = 6.37$$

$$P(Z=T|W=T,S=T,N=T,B=F) = 0.09$$

$$P(Z=F|W=T,S=T,N=T,B=F) = 0.91$$

Question 2

We seek to classify documents as being about sports or not. Each document is associated with a tuple (x,y) , where x is bag of word (BoW) representation of the document and y is the label for whether it is about sports ($y=1$ if yes, $y=0$ if false). The dictionary/vocabulary consists of 3 words (w_1, w_2, w_3), so BoW representation looks like $(1,3,0)$, $(1,4,5)$ etc, where the integers count the number of times the three words of the dictionary appears in the document.

Consider the naïve Bayes model with the following conditional probability table

y	$P(y)$
1	4/10
0	6/10

y	$P(w=w_1 y)$	$P(w=w_2 y)$	$P(w=w_3 y)$
1	1/10	2/10	7/10
0	5/10	2/10	3/10

Q2

Consider a document with BoW representation $x = (2,0,1)$. What is the posterior probability that the document is about sports.?

$$\begin{aligned} P(Y = 1 | x) &= a * P(Y=1) P(w_1 | y=1) P(w_2 | y=1) P(w_3 | y=1) \\ &= a * 4/10 * 1/10 * 1/10 * 7/10 \end{aligned}$$

$$\begin{aligned} P(Y = 0 | x) &= a * P(Y=0) P(w_1 | y=0) P(w_2 | y=0) P(w_3 | y=0) \\ &= a * 6/10 * 5/10 * 5/10 * 3/10 \end{aligned}$$

After normalization

$$P(Y = 1 | x) = .06$$

$$P(Y = 0 | x) = .94$$

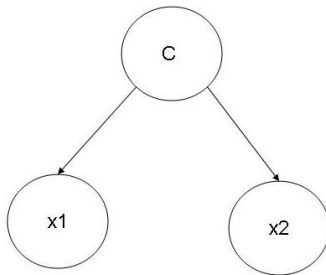
Q2

What is the probability (under naïve Bayes model) of seeing a document with BoW representation $x = (0,1,1)$, if we know that the document is about sports?

$$\begin{aligned} P(w_2, w_3 | Y = 1) &= P(w_2, w_3, Y=1) / P(Y=1) \\ &= P(w_2 | Y=1) P(w_3 | Y=1) P(Y=1) / P(Y=1) \\ &= P(w_2 | Y=1) P(w_3 | Y=1) = 2/10 * 7/10 = 14/100 = .14 \end{aligned}$$

Question 3

Below is the Bayes net structure for a Naive Bayes classifier, where C is the class (1 or 2) and x_1 and x_2 are binary features.



Use the following Conditional probability tables to classify the following observation according to the naive Bayes classifier

$$P(C = 1) = 0.6$$

C	$P(x_1 C)$
1	0.7
2	0.6

C	$P(x_2 C)$
1	0.5
2	0.8

Find the probability of $P(C | x_1=2 \text{ and } x_2=2)$